

Manual

Machining Center/ Pool of Orders BDE-BEA 8.1

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1 Overview of Machining Center/Pool of Orders

Possible fields of application

This component allows for the problems of a "machining center/pool of orders" to be represented.

Implementation notes

This function package is used if you have machines or machining centers where different items/orders are processed simultaneously.

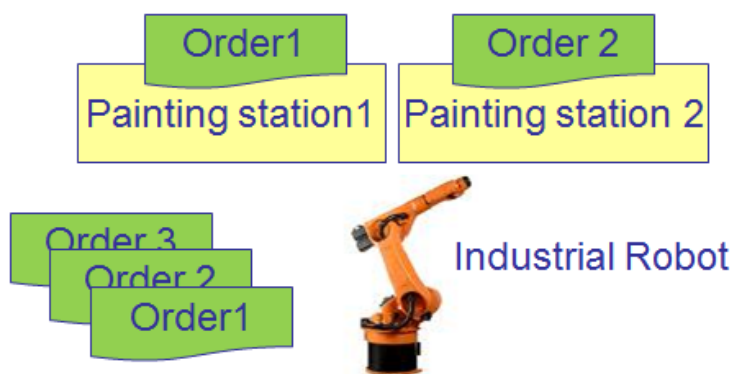
In a machining center several operations are logged on to a machine (= machining center) at the same time. As several operations are processed sequentially at a machine, they need to be differentiated to post the duration and labor utilization only to the active operation. In HYDRA-MDE (Machine Data Collection) produced quantities are only posted onto active operations.

In the classical sense of a machining center, e.g. a robot works on several tables with different orders at the same time. In a machining center several orders can be processed alternately at a machine or workstation.

This example is to explain how machining centers can be used. A machining center has two painting stations that are fed by a robot. Painting station 1 uses the color blue and painting station 2 uses the color green. Different articles are produced (item 1 and item 2). One operation or one production order is required for each article/item to be produced. This example has two production orders:

	Blue	Green
Article	Article 1	Article 2

These two articles can be produced and painted at the same time. The items can be processed in different order or with different quantities at the robot.



Integration

Two different connection or collection variants are possible:

1. Interactive activation of an operation (from the list of running operations) at the terminal. Downtime monitoring is realized by machine interface/UMPS or manual notification.
2. Recording of the events "order start" and "order end" using an interface. Provided that machines are connected correspondingly, information can be received from the machine or machine control to automate the change of operations.

Functions

- Functions to map specific requirements when working with a pool of orders at machining centers
- Configuration of machines and workplaces as machining center including pool of orders
- Login of several operations as pool of orders and login of people to the Windows terminal
- Recording of the events order start and order end, i.e. posting of the actual time when processing of an operation has started or ended
- Automatic registration of events by data taken over from machine controls (SPS) using an interface or manual registration by inputting events at the shop floor terminal
- Downtime monitoring at the Windows terminal including alternative downtime monitoring using machine interfaces or manual input
- Direct posting of recorded quantities and production times onto the corresponding operation
- Proportionate posting of downtimes and malfunction periods onto the registered operations

2 Machining Center / Order Pool

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Usage

The "Machining center / Order pool" functionality in AIP allows for the sequential processing of several operations logged on at the same time.

Prerequisite

Using the functionality requires provision of the configurations described in section [Configuration](#) .

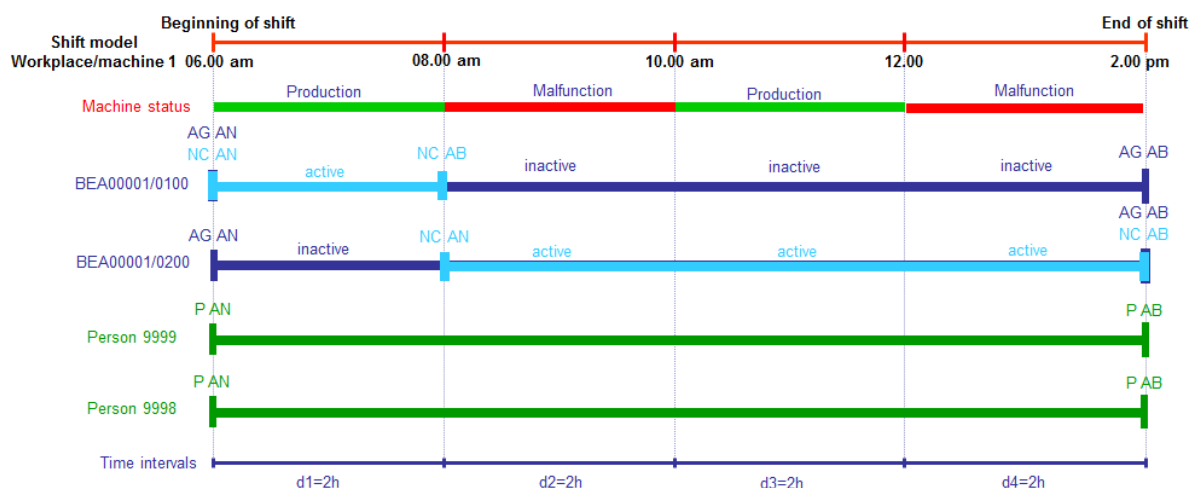
Logon Procedure and Posting

Different operations are logged on to the relevant machine by the operator. Other persons may log on optionally.

Precisely one operation is edited at a time. If the machine is in the "Production" mode, the duration and full labor utilization is posted to the active OP only. If no operation is active, no duration is recorded as regards the order. If there is a machine failure, the duration and labor utilization is proportionately posted on all operations.

Machining Center with Order Pool

Four operations are logged on to the machine at the same time!



Postings in HYDRA

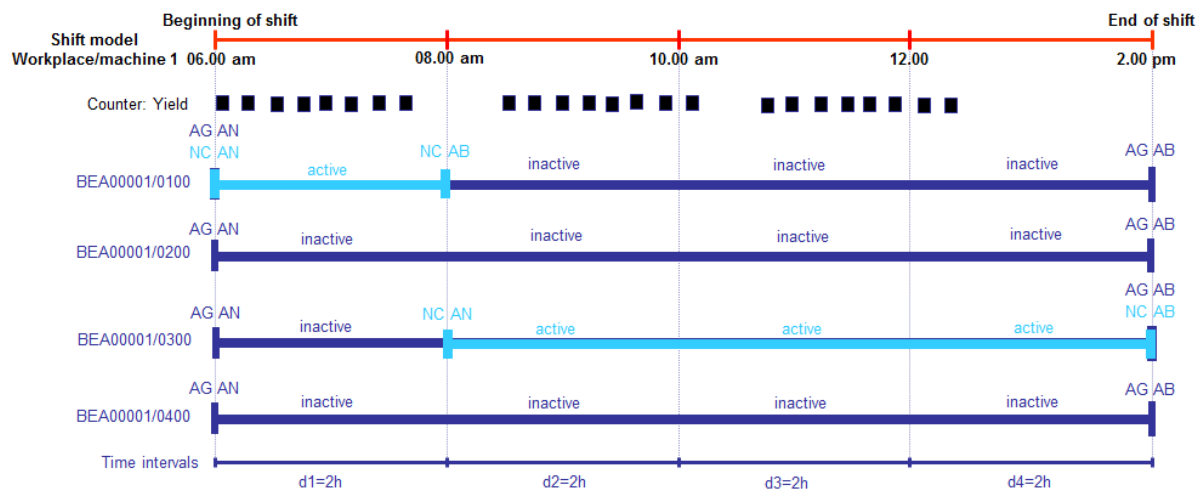
Order
OP BEA00001/0100 (U record)
06.00 - 14.00
⇒Duration: 2,0 hrs.
⇒Labor duration: 8,0 hours

Order
OP BEA00001/0300 (U record)
06.00 - 14.00
⇒Duration: 6,0 hrs.
⇒Labor duration: 8,0 hours

If automatic quantities are recorded for this machine, posting is performed in analogy to the time of production: the automatic quantities are only assigned to the active operation and the persons logged on to it. Manual quantities are assigned to the operation for which the quantities are posted.

Machining Center with Order Pool

Four operations are logged on to the machine at the same time!



Postings in HYDRA

Order OP BEA00001/0100 (U record) 06.00 - 08.00 ⇒Duration: 2,0 hrs. ⇒Quantity: 8 items	Order OP BEA00001/0200 (U record) 08.00 - 14.00 ⇒Duration: 0,0 hrs. ⇒Quantity: 0 items	Order OP BEA00001/0300 (U record) 08.00 - 14.00 ⇒Duration: 6,0 hrs. ⇒Quantity: 16 items	Order OP BEA00001/0400 (U record) 08.00 - 14.00 ⇒Duration: 0,0 hrs. ⇒Quantity: 0 items
--	--	---	--

A HYDRA group workplace, too, may represent a machining center. In this regard, however, automatic connection via interface is not possible.

The formation of a merged operation at a machining center is not permissible, since production and disturbance times are always posted proportionately for merged operations and this is in contradiction to the definitions of a machining center.



The posting events "Activate operation" and "Deactivate operation" are displayed in the event maintenance (Events NC_AN, NC_AB).

Function at Terminal

The "Operations at workplace" list in the basic screen shows the operations logged on. Given the relevant configuration of the list of orders (cf. section [Configuration](#)), the relevant active operation is color-marked.

In the "Activate OP" dialog, all operations logged on to the machine are shown (including the currently active operation). The operation marked in the basic screen is adopted as the selection.

In the "Deactivate OP" dialog, the active operation is set automatically.

Automatic deactivation of the active operation

Parameter in ctaip.ini → [System] → NC_AUTO_DEACTIVATE=Y (Default: N)

This button deactivates the validation function that only one OP may be active, and an automatic process is activated instead: the currently (previously) active operation is deactivated first and the new operation is activated subsequently. It is to be observed that this process is not related to a transaction, but that deactivation takes place even if subsequent activation is not possible due to an error. Such processing is effective both with manual operation on the terminal and if controlled via a [PCC-ADP telegram](#).

Configuration

Activation on MOC

The "Machining center" functionality is activated in the Machine / Workplace configuration :

Menu *Master data* → *Workplace/Resource configuration* → *Tab Workplace configuration* → *Workplace category=J <Machining center>*

Display of the active operation at the terminal

In order to highlight the currently active operation at the terminal, the settings listed below are required in ctaip.ini:

```
[Order list]
...
GRID_COLOR=clSilver
GRID_BACKGROUND=clWhite
EXAMINE_SCANEXPR1=AKTIV=J
EXAMINE_SCANEXPR2=AKTIV=N
EXAMINE_SCANCOLOR1=clBlack
EXAMINE_SCANBKEXPR1=AKTIV=J
EXAMINE_SCANBKEXPR2=AKTIV=N
EXAMINE_SCANBKCOLOR1=clLime
EXAMINE_SCANBKCOLOR2=clWhite
....
```

Configuration of keys in ctaipbut.ini

Buttons may be freely positioned, but consideration of the existing buttons is required. It must also be observed that the buttons are entered in the requested section and numbers are consecutive. Example:

```
[ANR-ALL-Page4]
1=NC_AN,L,Activate OP
2=NC_AB,L,Deactivate OP
```


Connection via PCC-ADP Telegram



Connection via PCC-ADP requires additional licenses.

The details of the planned connection are coordinated through joint discussions and the required settings on the application adapter are made in the course of customizing.

For activating the processing of a machine, the "Machining center" value is to be set active as the *Type* in the machine configuration (see above). For these machines, the terminal evaluates the data sent via a PCC-ADP telegram. Operations must previously be logged on at the terminal manually.

An automatic status change and/or downtime/standstill monitoring as for the file interface will not take place!

Dialog string for entry

Field identification	Type	Description
DLG	C10	Dialog/Record identification: "NC_AN" – Activate OP "NC_AB" – Deactivate OP
MNR	C20	Workplace/Machine number according to configuration
ANR	C20	MES order number (order/OP)

Should identification of the operation not be possible in the PCC-ADP telegram, a customer-specific extension is required in order to supplement the dialog string by the operation number!